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MAST20030: Differential Equations
Tutorial: Week 4

Question 1

(a) Find the Taylor series for $\log(1-t)$ about $t=0$. What is its radius of convergence?

(b) Find a first-order initial value problem for which $\log(1-t)$ is the unique solution. Show that the Taylor series you constructed in (a) solves this IVP, hence that the series converges to $\log(1-t)$ for all t in the interval of convergence.
Hint: you may use the result of a geometric series: $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$, for $|r|<1$ and $a \in \mathbb{R}$.

Question 2

In this question, we shall find two linearly independent solutions to the following second-order linear ODE:

$$y'' - ty' - y = 0.$$

(a) Substitute a general Taylor series about $t=0$, $y = \sum_{n=0}^{\infty} c_n t^n$, into the ODE and deduce a relation between a given c_n and c_{n+2}.

(b) Solve this relation so that c_{2m} depends explicitly on c_0 and c_{2m+1} depends explicitly on c_1.

(c) Write down two linearly independent solutions and determine their radii of convergence.

Question 3 (Uniform convergence)

In both questions 1 and 2, we relied on the fact we could bring the differential operator inside the infinite summation. Using the theorem's of 'Weierstrass test for uniform convergence' and 'Differentiating a series', demonstrate that in both questions, we were justified in bringing the differential operator inside the infinite summation.

Question 4 (Uniform convergence of a power series)

Theorem 1. *Every power series converges uniformly on every closed interval that lies strictly within its interval of convergence.*

Prove Theorem 1 for a power series about $t=0$

$$\sum_{n=0}^{\infty} c_n t^n,$$

with radius of convergence $r>0$.